

Random Variables, X

↳ A function that assigns a numerical value to each outcome in a sample space of a random experiment.

↳ Types of Random Variables:



Discrete

↳ Countable number of distinct values

↳ Example: # of heads in n coin flips

Continuous

↳ Uncountable numbers within an interval

↳ Example: heights of people within six feet

◆ 1. The Random Variable is the Observation

- This is the simplest relationship.
- The outcome of a random experiment is directly recorded as the value of the random variable.

✓ Example:

You toss a coin:

- Observation: "Head"
- Define random variable X such that:
 - $X = 1$ if Head
 - $X = 0$ if Tail

So the random variable directly represents the observed outcome.

◆ 3. The Random Variable is a Function of Another Random Variable

- You define one random variable based on another random variable.
- This is common in **transformations, derived distributions, or conditional probability models.**

✓ Example:

Let X = number of heads in 3 coin tosses.

Define $Y = 2X + 1$

- Here, Y is a function of another random variable X .
- If $X = 2$ (i.e., 2 heads), then $Y = 2(2) + 1 = 5$

◆ 2. The Random Variable is a Function of the Observation

- The **observation is more complex**, and the random variable is derived by applying a function to that observation.

✓ Example:

You roll a pair of dice:

- Observation: (3, 4)
- Random variable X = sum of dice = $3 + 4 = 7$

Here, the random variable is a function of the observation:

$$X = f(\text{observation}) = f(3, 4) = 7$$

✦ Summary Table:

Type	Description	Example
1. Random variable is the observation	Direct mapping	Toss coin $\rightarrow X = 1$ (Head), 0 (Tail)
2. Function of observation	Derived from raw data	Dice roll (2,5) $\rightarrow X = \text{sum} = 7$
3. Function of another random variable	Transformed variable	$X = \text{heads in tosses}$, $Y = 2X + 1$



In summary, a random variable is a numerical representation of outcomes in a random experiment.

Basic Example: Flip a coin 4 times

∴ $\mathcal{S} = \{ \text{HHHH}, \text{HHHT}, \text{HHTT}, \dots, \text{T T T T} \}$

if, $X = \#$ heads that occur.

$$X \in \{0, 1, 2, 3, 4\}$$

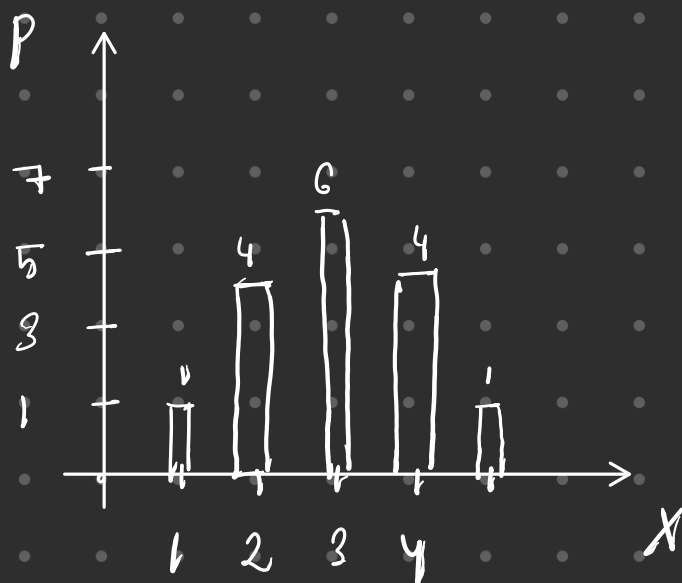
$$P(X=0) = \frac{1}{16}$$

$$P(X=1) = \frac{4}{16}$$

$$P(X=2) = \frac{6}{16}$$

$$P(X=3) = \frac{4}{16}$$

$$P(X=4) = \frac{1}{16}$$



So, the formal definition of a random variable $P(X)$ is a function called a distribution that contains

↳ values of X

↳ Probability P of each value

✱ It gives a concise summary of all probabilities and helps to visualize and compute.

PMF - Probability Mass Function

↳ Used in Discrete Random Variables

↳ Gives the probability that a discrete random variable is exactly equal to a value.

⊗ If X is a discrete random variable, then PMS is $P(X=x)$
This gives the probability that the random variable equals x .

↳ For example: Toss a fair coin once:

Let X = number of heads. | ⊗ $P(X=1) = 0.5$
⊗ $P(X=0) = 0.5$

PDF - Probability Density Function (will be discussed in next chapter)

↳ Used in Continuous Random Variables.

↳ Gives the density of the probability at a point.
It doesn't give the actual probability of a value.
The probability is computed over an interval using integration

⊗ If X is a continuous random variable:
 $P(a \leq X \leq b) = \int_a^b f(x) dx$; $f(x) \rightarrow$ PDF

↳ For example: Let $X \sim U(0,1)$ (uniform distribution from 0 to 1) then: $f(x) = 1$ for $0 \leq x \leq 1$ || $P(0.2 \leq x \leq 0.6) = \int_{0.2}^{0.6} 1 dx = 0.4$

* For a discrete random variable X with PMF $P_X(x)$ and range S_X .

(a) For any x , $P_X(x) \geq 0$ [Non-negativity]

(b) $\sum_{x \in S_X} P(x) = 1$. [Normalization]

(c) For any event $B \subset S_X$, the probability that X is in the set B is $P[B] = \sum_{x \in B} P_X(x)$ [Additivity for events]

The random variable N has PMF

$$P_N(n) = \begin{cases} c/n & n = 1, 2, 3, \\ 0 & \text{otherwise.} \end{cases}$$

Find

(a) The value of the constant c

(b) $P[N = 1]$

(c) $P[N \geq 2]$

(d) $P[N > 3]$

Example:

Solⁿ:

(a) $P_N(n)$ is a PMF.

$$\text{Here, } \sum_{n \in S_N} P(n) = 1$$

$$\rightarrow \sum_{n=1}^3 P_N(n) = 1 \Rightarrow \frac{c}{1} + \frac{c}{2} + \frac{c}{3} = 1 \Rightarrow C = \frac{6}{11}$$

$$(b) \text{ If, } P[N=1] = \frac{C}{1} = \frac{6}{11} \\ \therefore P[N=1] = \frac{6}{11}$$

$N \rightarrow$ random variable
 $n \rightarrow$ specific value N takes.

$$\textcircled{c} \quad P[N \geq 2] = P[N=2] + P[N=3] = \frac{C}{2} + \frac{C}{3}$$

$$= \frac{6}{11} \left(\frac{1}{2} + \frac{1}{3} \right)$$

$$= \frac{5}{11}$$

$\textcircled{d} \quad P[N > 3] = 0$; PMF is defined only for $n = 1, 2, 3, 0$

Example from Slide:

Example 2.1: (Continued)

Procedure: Send 3 packets from a sender to a receiver.

Observation: Number of successes.

$S = \{FFF, FFD, FDF, FDD, DFF, DFD, DDF, DDD\}$

$E = \{E_0, E_1, E_2, E_3\}$

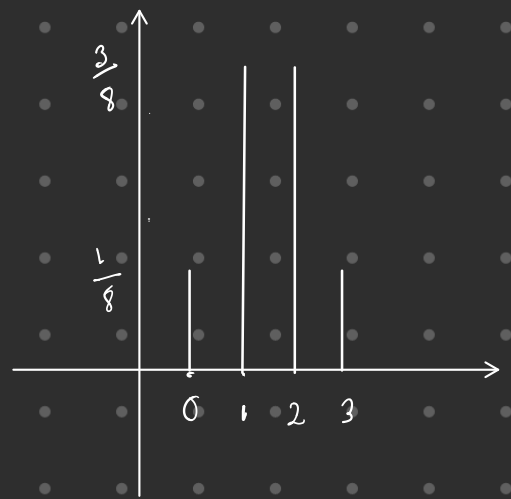
$X \triangleq$ Random variable that counts the number of successes

$S_X = \{0, 1, 2, 3\}$

n (successes)	E_x	s (sequences)	$P[X=n]$	$P[X \leq n]$
0	E_0	DDD	$1/8$	$1/8$
1	E_1	FDD, DFD, DDF	$3/8$	$1/8 + 3/8$
2	E_2	FFD, FDF, DFF	$3/8$	$1/8 + 3/8 + 3/8$
3	E_3	FFF	$1/8$	$1/8 + 3/8 + 3/8 + 1/8$

So, the PMF would be:

$$P_X(x) = \begin{cases} 1/8 & \text{if } x = 0, 3 \\ 3/8 & \text{if } x = 1, 2 \\ 0 & \text{otherwise} \end{cases}$$



⊗ Falls onto Binomial Distribution Technique.
This will be shown later on.

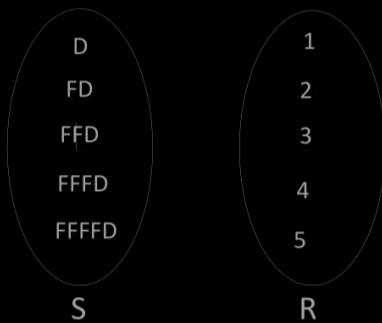
Example from Slides

Example 2.2 (Example 1.13 Continued)

Procedure: Keep sending packets from a sender to a receiver until 1 packet is delivered

Observation: Number of attempts

$$S = \{D, FD, FFD, FFFD, \dots\} \quad s_X = \{1, 2, 3, \dots\}$$



X = number of attempts until success.

Possible values: $s_X = \{1, 2, 3, \dots\}$

$$p(\text{success}) = p = 0.5$$

$$p(\text{failure}) = 1 - p = 0.5$$

	E_1	E_2	E_3	E_4	E_5	E_6
S	D	FD	FFD	FFFD	FFFFD	FFFFFD
x	1	2	3	4	5	6
$P[X=x]$	0.5	0.5^2	0.5^3	0.5^4	0.5^5	0.5^6
$P[X \leq x]$	0.5	$0.5 + 0.5^2$	$0.5 + 0.5^2 + 0.5^3$	$0.5 + 0.5^2 + 0.5^3 + 0.5^4$	$0.5 + 0.5^2 + 0.5^3 + 0.5^4 + 0.5^5$	$0.5 + 0.5^2 + 0.5^3 + 0.5^4 + 0.5^5 + 0.5^6$

$$\therefore \text{PMF} : P(X=x) = \begin{cases} (0.5)^x & \text{for } x=1, 2, 3, \dots \\ 0 & \text{otherwise} \end{cases}$$

⊛ Falls onto Geometric Distribution Technique.
This will be shown later on.

Example

from Slide:

X = number of packets until three successes

$$P(\text{success}) = p = 0.5$$

$r = 3$ (number of success we want)

Example 2.3: (Example 1.14 Continued)

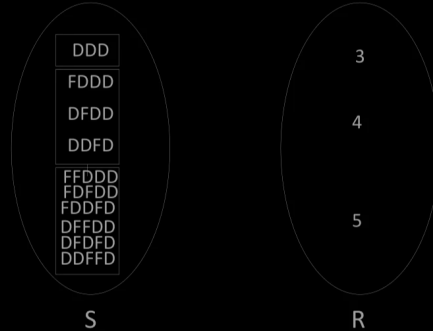
Example 1.14 (Experiment 1.5)

Procedure: Keep sending packets from a sender to a receiver until 3 packets are delivered

Observation: Number of attempts

$$S = \{DDD, FDDD, DFDD, DDFD, FFDDD, \dots\}$$

$$S_X = \{3, 4, 5, \dots\}$$



S

x

$$P[X=x]$$

$$P[X \leq x]$$

E_3

DDD

3

$$p^3$$

$$p^3$$

E_4

FDDD, DFDD,
DDFD

4

$$3 \cdot p^3(1-p)$$

$$3p^3(1-p) + p^3$$

E_5

FFDDD, FDFDD,
FDDFD, DFFDD,
DFFDD, DFFDD.

5

$$6 \cdot p^3(1-p)^2$$

$$p^3 + 3p^3(1-p) + 6p^3(1-p)^2$$

$$\therefore \text{PMF} : P(X=x) = \begin{cases} \binom{x-1}{r-1} p^r (1-p)^{x-r}, & x=3,4,5,\dots \\ 0 & \text{otherwise} \end{cases}$$

* Falls onto Pascal Distribution Technique
This will be shown later on.

CDF - Cumulative Distribution Function

↳ denoted by $F_X(x)$, gives the probability that X takes a value less than or equal to x ,

$$F_X(x) = P(X \leq x)$$

This accumulates the probabilities of all outcomes up to and including x . So, for any real number x , the CDF is the probability that the random variable X is no larger than x .

If X is discrete, the CDF is the sum of the PMF values up to x :

$$F_X(x) = \sum_{k=-\infty}^{\lfloor x \rfloor} p_X(k)$$

⊗ All random variables have cumulative distribution functions but only discrete random variables have probability mass functions.

Properties of CDF:

① Non-decreasing: CDF always increases or stays constant. For all $x' \geq x$, $F_X(x') \geq F_X(x)$.

② $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow \infty} F_X(x) = 1$

Can be used to compute probability ranges:

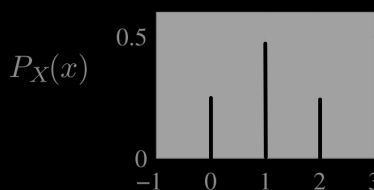
$$P(a < X \leq b) = F_X(b) - F_X(a)$$

(iv) $F_X(x) = F_X(x_i)$ for all x such that $x_i \leq x < x_{i+1}$

Example:

Example 3.21

In Example 3.5, random variable X has PMF



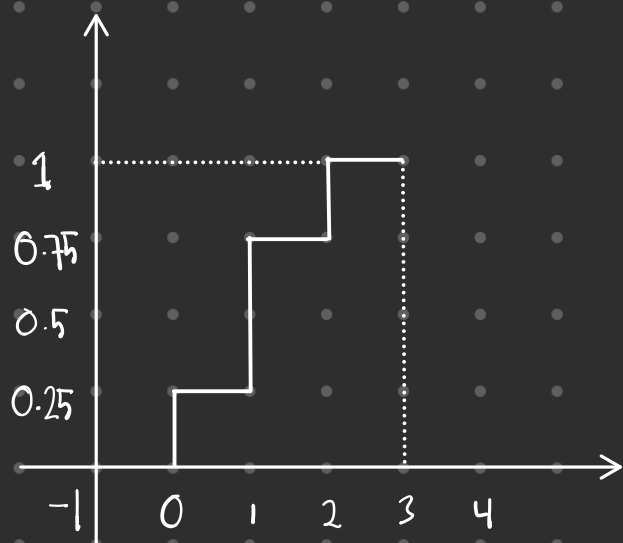
$$P_X(x) = \begin{cases} 1/4 & x = 0, \\ 1/2 & x = 1, \\ 1/4 & x = 2, \\ 0 & \text{otherwise.} \end{cases} \quad (3.28)$$

Find and sketch the CDF of random variable X .

Soln:

$$F_X(x) = P[X \leq x]$$

$$= \begin{cases} 0 & x < 0 \\ 1/4 & 0 \leq x < 1 \\ 3/4 & 1 \leq x < 2 \\ 1 & x \geq 2 \end{cases}$$



Example from the slide:

Example 2.4: Let a discrete random variable X assumes values $-1, 1, 2$, and 3 , with probabilities $0.6, 0.3, 0.08$, and 0.02 , respectively.

$$\text{PMF: } P_X(x) = \begin{cases} 0.6, & x = -1 \\ 0.3, & x = 1 \\ 0.008, & x = 2 \\ 0.002, & x = 3 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{CDF: } F_X(x) = \begin{cases} 0, & x \leq -1 \\ 0.6, & -1 \leq x < 1 \\ 0.9, & 1 \leq x < 2 \\ 0.98, & 2 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

Average and Expected Value

An average is a number that describes a set of experimental observations. In probability, this is called the expected value or mean, denoted as: $E[X]$.

The expected value of X is

$$E[X] = \mu_X = \sum_{x \in S_X} x p_X(x)$$

For Example: A die is rolled. The random variable X is the number on the face.

Here, $p_X(x) = 1/6$ for $x = 1, 2, 3, 4, 5, 6$

$$E[X] = \sum_{x=1}^6 x \cdot \frac{1}{6} = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5$$

Even though 3.5 is not a possible outcome it's the average of many rolls.

Example: Suppose you play a game, where in each play you lose \$1 with probability 0.60 and you win \$1 \$2 and \$3 with probabilities 0.30 0.08 and 0.02, respectively. Find your net gain in each play of the game.

-If you play only a few times, your gain might depend on your luck

- If you play a large number of times - your luck no longer determines the total gain - rather it will be multiple of average gain

Say, you play the game n times, approximately in $0.6 \times n$ games you will loose \$1 per game and will win \$1, \$2 and \$3 in $0.3 \times n$, $0.08 \times n$ and $0.02 \times n$ games, respectively.

x	-1	1	2	3	Total ↓
$P_X(x)$	0.60	0.30	0.08	0.02	1.00
$x \cdot P_X(x)$	-0.60	0.30	0.16	0.06	-0.08

$$P_X(x) = \begin{cases} 0.60 & x = -1 \\ 0.30 & x = 1 \\ 0.08 & x = 2 \\ 0.02 & x = 3 \\ 0 & \text{o/w} \end{cases}$$

$$\text{Here, Total gain} = .60n(-1) + .30n(1) + .08n(2) + .02n(3) = -0.08n$$

$$\therefore \text{Expected Value} = \frac{-0.08n}{n} = -0.08$$

Example: Three packets are sent from one computer to another. Each packet is successfully sent with a probability of 0.8 irrespective of the others. Each successfully sent packet results in a reward of 1 unit whereas 1 unit penalty (negative reward) is imposed if none of the packets is sent successfully. Find the expected net reward.

Solⁿ: Possible values $S_x = \{-1, 1, 2, 3\}$

$$P(-1) = q^3 = 0.008$$

$$P(1) = 3pq^2 = 0.096$$

$$P(2) = 3p^2q = 0.384$$

$$P(3) = p^3 = 0.512$$

$$P_X(x) = \begin{cases} 0.008 & x = -1 \\ 0.096 & x = 1 \\ 0.384 & x = 2 \\ 0.512 & x = 3 \\ 0 & \text{o/w} \end{cases}$$

Three Steps:

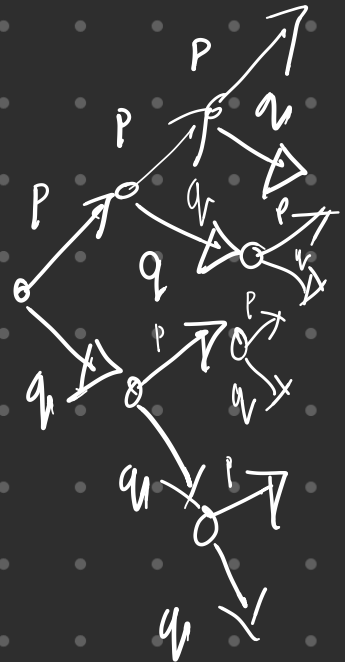
1. Find Random Variable

2. Find PMF.

3. Finally find $E[X]$

$$p = 0.8$$

$$q = 0.2$$

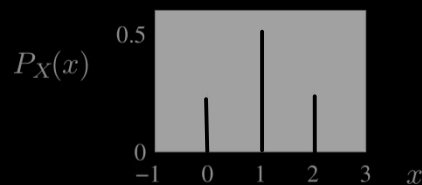


$$\therefore E[X] = (-1) \times 0.008 + 1 \times 0.096 + 2 \times 0.384 + 3 \times 0.512$$

$$= 2.32$$

Example 3.24

Random variable X in Example 3.5 has PMF



$$P_X(x) = \begin{cases} 1/4 & x = 0, \\ 1/2 & x = 1, \\ 1/4 & x = 2, \\ 0 & \text{otherwise.} \end{cases} \quad (3.36)$$

What is $E[X]$?

Example:

Soln:

$$\begin{aligned} \text{Here, } E[X] &= \left(\frac{1}{4} \times 0\right) + \left(\frac{1}{2} \times 1\right) + \left(\frac{1}{4} \times 2\right) \\ &= 1. \end{aligned}$$

Distribution

Expected values, $E[X]$

Bernoulli (p)

p

Binomial (n, p)

np

Geometric (p)

$1/p$

Discrete Uniform (k, l)

$(k+l)/2$

Pascal (k, p)

k/p

Poisson (α)

α

⊛ proof can be done using PMF.

Examples:

Solⁿ:

Quiz 3.5

In a pay-as-you go cellphone plan, the cost of sending an SMS text message is 10 cents and the cost of receiving a text is 5 cents. For a certain subscriber, the probability of sending a text is $1/3$ and the probability of receiving a text is $2/3$. Let C equal the cost (in cents) of one text message and find

- (a) The PMF $P_C(c)$
- (b) The expected value $E[C]$
- (c) The probability that the subscriber receives four texts before sending a text.
- (d) The expected number of texts received by the subscriber before the subscriber sends a text.

(a) the PMF is: $P_C(c) = \begin{cases} 1/3, & \text{if } c = 10 \\ 2/3, & \text{if } c = 5 \\ 0, & \text{otherwise} \end{cases}$

(b) $E[C] = \sum_C C \cdot P_C(c)$
 $= 10 \cdot \frac{1}{3} + 5 \cdot \frac{2}{3} + 0 = \frac{20}{3} = 6.67 \text{ cents}$

(c) $p(4 \text{ receives, then 1 send}) = \left(\frac{2}{3}\right)^4 \cdot \left(\frac{1}{3}\right)$
 $= \frac{16}{243} \approx 0.0658$

Example: A college mathematics department sends 8 to 12 professors to the annual meeting of the American Mathematical Society, which lasts five days. The hotel at which the conference is held offers a bargain rate of a dollars per day per person if reservations are made 45 or more days in advance, but charges a cancellation fee of $2a$ dollars per person. The department is not certain how many professors will go. However, from past experience it is known that the probability of the attendance of i professors is $1/5$ for $i = 8, 9, 10, 11, \text{ and } 12$. If the regular rate of the hotel is $2a$ dollars per day per person, should the department make any reservations? If so, how many?

Solⁿ:

$$S_X = \{8, 9, 10, 11, 12\}$$

i = professors

r = reservations

Three possible Cases:

Case 1: if all reservations are used as planned
($i=r$) Cost = $5ar$

Case 2: if extra rooms ($i-r$) are needed.
Cost = $5(r a + (i-r) 2a)$

Case 3: if only i rooms are used, the rest ($r-i$) rooms are cancelled.
Cost = $5ia + (r-i) 2a$.

Using the formulas, find expected values:

$$E(X_8) = \frac{1}{5} (40a + 50a + 60a + 70a + 80a) = 60a$$

$$E(X_9) = \frac{1}{5} (42a + 45a + 55a + 65a + 75a) = 56.4a$$

$$E(X_p) = \frac{1}{5} (44a + 47a + 50a + 60a + 70a) = 54.2a$$

$$E(X_{11}) = \frac{1}{5} (46a + 49a + 52a + 55a + 65a) = 53.4a$$

$$E(X_{14}) = \frac{1}{5} (48a + 51a + 54a + 57a + 60a) = 54a$$

We can see that X_{11} has the smallest expected value. Thus making 11 reservations is the most reasonable policy.

Derived Random Variable

↳ A derived random variable is formed by applying a function to another random variable.

Let X be a random variable, and let $Y = g(X)$, where g is some function. Then Y is called a derived or transformed random variable.

For Example:

$$P_X(x) = \begin{cases} 0.60 & x = -1 \\ 0.30 & x = 1 \\ 0.08 & x = 2 \\ 0.02 & x = 3 \\ 0 & \text{o/w} \end{cases}$$

and if $Y = X^2$

What will be the PMF of Y ?

Solⁿ:

Here, $S_X = \{-1, 1, 2, 3\}$

So if, $Y = X^2 \rightarrow S_Y = \{1, 4, 9\}$

$$\therefore P_Y(y) = \begin{cases} .60 & y = 1 \\ .08 & y = 4 \\ .02 & y = 9 \end{cases}$$

$$P[Y=1] = P[X=-1] +$$

$$P[X=1]$$

$$= .60$$

$$P[Y=4] = P[X=2]$$

$$= .08$$

$$P[Y=9] = P[X=3]$$

$$= .02$$

⊛ For a discrete random variable Y , if there exists another random variable X , where $Y = g(X)$, then the PMF of X is

$$P_Y(y) = \sum_{x: g(x)=y} P_X(x)$$

Expected value of a Derived Random Variable

$$\hookrightarrow E[Y] = E[g(x)] = \sum_x g(x) \cdot p_X(x)$$

For Example:

$$p_X(x) = \begin{cases} 0.60 & x=-1 \\ 0.30 & x=1 \\ 0.08 & x=2 \\ 0.02 & x=3 \\ 0 & \text{o/w} \end{cases} \quad \text{and} \quad Y = X^2$$

$$\therefore E[Y] = (-1)^2 \cdot 0.60 + (1)^2 \cdot 0.30 + (2)^2 \cdot 0.08 + (3)^2 \cdot 0.02 + 0 \\ = 1.4$$

$$\textcircled{*} E[X - \mu_X] = 0$$

$$\textcircled{*} E[X + Y] = E[X] + E[Y]$$

$$\textcircled{*} E[aX + b] = aE[X] + b$$

N^{th} Moment of a Random Variable

$$\hookrightarrow E[X^n] = \sum_x x^n \cdot p_X(x)$$

$$\text{So, if } n=0 \rightarrow E[X^0] = 1$$

$$n=1 \rightarrow E[X] \Rightarrow \text{expected value}$$

$$n=2 \rightarrow E[X^2] \Rightarrow \text{second moment}$$

$$= \sum_x x^2 p_X(x)$$

Variance

↳ A measure of how much a random variable deviates from its mean, on average - it quantifies spread or variability.

The variance is defined as the 2nd central moment:

$$\text{Var}(X) = E[(X - \mu)^2] \quad \text{where, } \mu = E[X]$$

$$\text{or, } \text{Var}(X) = E[X^2] - (E[X])^2$$

Standard Deviation

↳ the square root of the variance - it measures the average distance of the values from the mean, in original units of the data. It directly tells you how spread out the values are from the mean.

$$\text{SD}(X) = \sqrt{\text{Var}(X)} = \sqrt{E[(X - \mu)^2]}$$

Nth Central Moment $\rightarrow E[(X - \mu)^n]$

$$\text{if, } n=0 \rightarrow E[(X - \mu)^0] = 1$$

$$\text{if, } n=1 \rightarrow E[(X - \mu)^1] = 0$$

$$\text{if } n=2 \rightarrow E[(X - \mu)^2] = V[X] \rightarrow \text{Variance}$$

$$\text{if } n=3 \rightarrow E[(X - \mu)^3] \rightarrow \text{Skewness}$$

$$\text{if } n=4 \rightarrow E[(X - \mu)^4] \rightarrow \text{Kurtosis}$$